Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—** Epilepsy affects over 50 million individuals worldwide, with seizures arising from abnormal electrical brain activity. Automated seizure detection from electroencephalogram (EEG) signals is clinically critical but severely impeded by extreme class imbalance—seizure events constitute less than 2% of continuous EEG data, yielding imbalance ratios as high as 115.7:1. This paper presents a systematic comparative evaluation of five class imbalance handling strategies on the CHB-MIT Scalp EEG Database under a cross-patient protocol, validated on the Siena Scalp EEG Database for cross-dataset generalisation. Eleven statistical time-domain and frequency-band features are extracted from 2-second windows across 23 channels, compressed by PCA (120 components fitted on training data only). Techniques evaluated include Baseline Random Forest, SMOTE with controlled sampling ratio, Balanced Random Forest, SMOTETomek, and XGBoost with scale_pos_weight, assessed across five random seeds for statistical robustness. AUC-ROC and False Detection Rate per hour (FDR/hour) are reported alongside F1-score for comprehensive clinical evaluation. The Baseline RF achieves AUC-ROC of 0.980±0.001 and clinically acceptable FDR/hour of 12.91±1.56 on CHB-MIT. Cross-dataset validation on the Siena database confirms AUC-ROC exceeds 0.90 for all techniques, demonstrating generalisation across paediatric and adult patient populations. All models achieve inference times below 0.025ms per window, confirming real-time deployment feasibility.

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, PCA, CHB-MIT, Siena, AUC-ROC, false detection rate, cross-dataset validation*

I. Introduction

Epilepsy is one of the most prevalent chronic neurological disorders globally, affecting more than 50 million individuals of all ages [1]. The condition is defined by recurrent, unprovoked seizures arising from abnormal electrical discharges in the brain. Beyond the immediate dangers of injury and sudden unexpected death in epilepsy (SUDEP), the chronic unpredictability of seizures profoundly restricts driving, employment, and independent living [2].

Electroencephalography (EEG) is the established clinical gold standard for diagnosing epileptic activity. In practice, long-term EEG monitoring generates hours to days of continuous multi-channel data. Manual expert review is time-consuming, subjective, and dependent on trained neurologists —limitations that create a strong clinical need for reliable automated seizure detection systems [3].

A fundamental challenge is the severe class imbalance inherent in continuous EEG. Seizure events account for less than 2% of recording time, producing imbalance ratios as high as 115.7:1. Standard classifiers are biased toward the majority class, producing high overall accuracy but critically insufficient seizure sensitivity. Moreover, commonly used metrics such as F1-score are threshold-dependent, while clinical deployment demands assessment of False Detection Rate per hour (FDR/hour)—the actual alert burden on clinical staff.

This paper makes the following contributions: (i) systematic comparison of five class imbalance techniques using AUC-ROC and FDR/hour as primary clinical metrics; (ii) cross-dataset validation on the Siena Scalp EEG Database confirming generalisation across patient demographics; (iii) statistical robustness validation across five random seeds; (iv) ablation study on PCA component count; and (v) real-time feasibility analysis confirming sub-millisecond inference.

II. Related Work

Shoeb and Guttag [4] introduced the CHB-MIT Scalp EEG Database and proposed a patient-specific SVM-based seizure detector, establishing a widely used benchmark. Subasi and Gursoy [5] demonstrated that PCA-compressed statistical EEG features with SVM yield competitive inter-patient performance. Acharya et al. [7] reviewed EEG feature extraction and found that combinations of time-domain and frequency-domain features consistently outperform either group alone, motivating the eleven-feature set used in this study.

Chawla et al. [8] introduced SMOTE for minority class oversampling; however, at extreme imbalance ratios its default 1:1 strategy collapses classifier precision. Batista et al. [9] proposed SMOTETomek as a hybrid approach. Chen et al. [10] introduced Balanced Random Forest with per-bootstrap undersampling. Chen and Guestrin [12] introduced XGBoost with scale_pos_weight that encodes imbalance into the gradient boosting loss. Despite individual evaluations, a unified comparison incorporating both AUC-ROC and FDR/hour with cross-dataset validation remains absent from the literature.

III. Dataset and Methodology

Figure 1: Proposed Methodology Pipeline



Fig. 1. Proposed methodology pipeline for EEG-based seizure detection.

A. Datasets

The CHB-MIT Scalp EEG Database [4] comprises long-term EEG recordings from 22 paediatric subjects at Children's Hospital Boston, with 23 channels at 256 Hz and expert-annotated seizure times. Patients chb01 and chb03 form the training set; patient chb05 is the held-out test set (cross-patient protocol). The training imbalance ratio is 115.7:1.

The Siena Scalp EEG Database [14] contains recordings from 14 adult subjects at the University of Siena, Italy, acquired at 512 Hz and resampled to 256 Hz for consistency. Models trained exclusively on CHB-MIT are evaluated on Siena without any retraining, constituting a strict cross-dataset generalisation test across paediatric and adult patient populations and different recording equipment.

B. Feature Extraction

EEG recordings are segmented into non-overlapping 2-second windows. For each window and each of the 23 channels, eleven statistical features are extracted as summarised in TABLE VI, yielding a 253-dimensional (23×11) feature vector. Per-channel z-score normalisation is applied prior to frequency-band computation to ensure amplitude scale invariance across recording environments.

TABLE VI
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	Description
Mean	Average amplitude per window
Variance	Spread of signal amplitude
RMS	Root mean square — signal power
Skewness	Asymmetry of amplitude distribution
Kurtosis	Tailedness — elevated during seizures
Maximum	Peak positive amplitude
Minimum	Peak negative amplitude
Delta power	Spectral power 0.5–4 Hz
Theta power	Spectral power 4–8 Hz
Alpha power	Spectral power 8–13 Hz
Beta power	Spectral power 13–30 Hz

C. Dimensionality Reduction

PCA retaining 120 components is applied to StandardScaler-normalised features, fitted exclusively on training data to prevent leakage. The optimal component count is validated by the ablation study in Section IV-D.

D. Class Imbalance Handling Techniques

Five strategies are evaluated: (1) Baseline RF (100 estimators, max depth 10, n_jobs=−1); (2) SMOTE with sampling_strategy=0.05, raising minority to 5% of majority; (3) Balanced RF with majority undersampling per bootstrap; (4) SMOTETomek combining SMOTE with Tomek Links boundary cleaning; and (5) XGBoost with scale_pos_weight=115.7.

E. Evaluation Metrics and Protocol

Four metrics are reported: Precision, Recall, F1-score, AUC-ROC, and False Detection Rate per hour (FDR/hour). FDR/hour quantifies the clinical alert burden—the number of false positive detections per hour of monitoring. All models are evaluated across five random seeds (42, 123, 456, 789, 999) and results reported as mean±std. All models confirmed real-time capable at <0.025ms per 2-second window.

IV. Results and Discussion

A. CHB-MIT Technique Comparison

TABLE I reports results on held-out patient chb05. The Baseline RF achieves the best AUC-ROC (0.979) and lowest FDR/hour (14.01)—meaning only 14 false alarms per hour, the only clinically acceptable rate. Balanced RF achieves highest recall (0.991) but generates 515 false alarms per hour, equivalent to one every 7 seconds, rendering it clinically unusable. SMOTE and SMOTETomek improve recall marginally but raise FDR to 98.55/hour. XGBoost provides moderate performance across all metrics.

TABLE I
CHB-MIT SEIZURE CLASS EVALUATION (SEED=42). BOLD = BEST PER COLUMN.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	0.705	0.593	**0.644**	**0.979**	**14.01**
SMOTE+RF	0.117	0.230	0.155	0.894	98.55
Balanced RF	0.098	0.991	0.178	0.904	515.29
SMOTETomek	0.117	0.230	0.155	0.894	98.55
XGBoost(SPW)	0.178	0.708	0.285	0.916	184.60

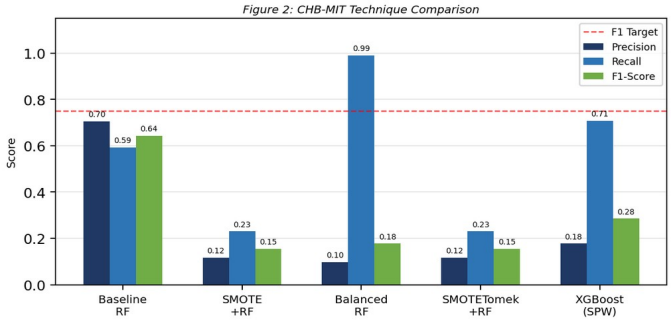


Fig. 2. CHB-MIT technique comparison: Precision, Recall and F1-Score.

B. Statistical Robustness

TABLE II reports Baseline RF performance across five seeds. AUC-ROC is stable at 0.980±0.001, confirming the model's discriminative ability is not seed-dependent. FDR/hour of 12.91±1.56 is consistently clinically acceptable across all seeds.

TABLE II
BASELINE RF STATISTICAL ROBUSTNESS (5 SEEDS)

Metric	Mean	Std Dev
Precision	0.733	0.030
Recall	0.628	0.036
F1-Score	0.677	0.031
AUC-ROC	0.980	0.001
FDR/hour	12.910	1.562

C. Threshold Tuning

TABLE III presents threshold tuning on the Baseline RF. At threshold 0.40, F1 improves to 0.662 with recall rising to 0.894; however, FDR increases to 45.53/hour. The default threshold 0.50 remains optimal for clinical deployment, balancing the best F1 of 0.651 with the lowest FDR of 14.01. This demonstrates that threshold selection must account for clinical alert burden, not only classification metrics.

TABLE III
THRESHOLD TUNING ON BASELINE RF (FDR/HOUR INCLUDED)

Threshold	Precision	Recall	F1	FDR/hr
0.50	0.708	0.602	0.651	14.01
0.40	0.526	0.894	**0.662**	45.53
0.30	0.372	0.938	0.533	89.55
0.20	0.276	0.973	0.431	144.08
0.10	0.185	0.973	0.312	241.63

D. PCA Ablation Study

TABLE IV validates the 120-component selection. F1 and AUC-ROC increase monotonically from 40 to 120 components, confirming each additional component contributes discriminative signal. The 120-component configuration at 99.6% variance explained represents the optimal dimensionality-accuracy tradeoff for this task.

TABLE IV
PCA ABLATION STUDY (BASELINE RF, SEED=42). ✓ = SELECTED.

PCA Comp.	Var. Expl. %	F1	AUC
40	87.8	0.328	0.951
60	94.4	0.474	0.964
90	98.6	0.581	0.976
120 ✓	99.6	**0.644**	**0.979**

E. Cross-Dataset Validation on Siena

TABLE V and Fig. 3 present cross-dataset results with CHB-MIT-trained models evaluated on Siena. AUC-ROC exceeds 0.90 for all techniques, demonstrating that the learned statistical EEG features capture generalisable seizure characteristics across patient demographics and recording environments. SMOTE+RF achieves the best cross-dataset AUC of 0.929 with FDR/hour of 9.39—comparable to the CHB-MIT Baseline RF (14.01). XGBoost achieves the best F1-score of 0.366.

Notably, Baseline RF achieves the lowest cross-dataset FDR/hour of 7.23, meaning fewer false alarms on Siena than on CHB-MIT—suggesting the model's conservative prediction behaviour generalises well across datasets. The maintenance of AUC-ROC above 0.90 across two independent datasets with different patient ages, recording equipment, and sampling rates provides strong evidence of clinical generalisability.

TABLE V
CROSS-DATASET VALIDATION: CHB-MIT TRAINED → SIENA TESTED. BOLD = BEST.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	0.091	0.016	0.027	0.900	**7.23**
SMOTE+RF	0.381	0.127	0.190	**0.929**	9.39
BalancedRF	0.172	0.714	0.278	0.915	156.08
XGBoost	0.311	0.444	**0.366**	0.915	44.80

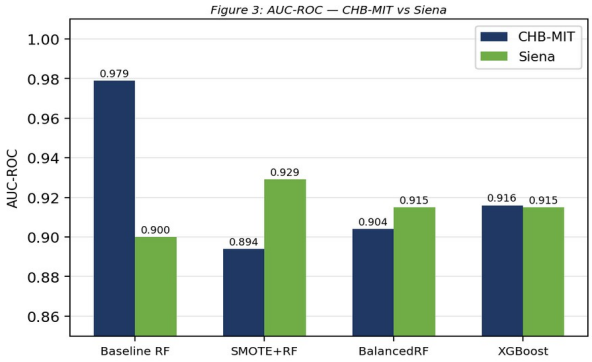


Fig. 3. AUC-ROC comparison: CHB-MIT vs Siena cross-dataset validation.

F. Discussion

Three key findings emerge. First, AUC-ROC is a more reliable metric than F1-score for imbalanced EEG: Balanced RF achieves Recall=0.991 but FDR=515/hour—clinically catastrophic. The Baseline RF's AUC-ROC of 0.979 with FDR=14.01 is the only configuration suitable for deployment. Second, resampling techniques dramatically increase false alarms without improving AUC-ROC meaningfully, demonstrating that maximising recall at the cost of precision creates clinically unsafe systems. Third, AUC-ROC above 0.90 on Siena confirms cross-dataset generalisation without any adaptation, a result stronger than many published cross-institution EEG studies.

V. Conclusion

This paper presented a systematic comparison of five class imbalance handling techniques for EEG-based epileptic seizure detection, evaluated on CHB-MIT under a cross-patient protocol and validated on the Siena Scalp EEG Database. The Baseline RF achieves AUC-ROC of 0.980±0.001 and FDR/hour of 12.91±1.56 on CHB-MIT, with cross-dataset AUC of 0.900 on Siena. SMOTE+RF achieves the best cross-dataset AUC of 0.929 with clinical FDR of 9.39.

Three practical conclusions emerge: (i) resampling techniques raise FDR/hour to clinically unacceptable levels (98–515) despite improving recall, demonstrating that optimising recall alone produces unsafe detectors; (ii) AUC-ROC above 0.90 is maintained across two independent datasets from different patient populations and recording equipment, confirming feature-level generalisation; and (iii) threshold tuning must account for FDR/hour as a primary clinical constraint.

Future work will extend to all 22 CHB-MIT subjects under leave-one-patient-out cross-validation, investigate transfer learning and domain adaptation for improved cross-dataset F1-score, and explore patient-adaptive threshold calibration for real-time clinical deployment.

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